

Group 6

Project Proposal: Enhanced Health Insurance Claims Analysis Using Big Data Tools

Problem Definition

The health insurance industry produces massive volumes of claims data daily, and efficient management of this data is critical for operational efficiency, fraud detection, and cost control. Traditional relational database systems struggle to manage such large datasets due to scalability and performance limitations. This leads to delays in detecting fraudulent activities, inaccurate predictions of claim costs, and limited ability to profile providers and patients effectively.

Our project aims to leverage big data management tools, specifically Hadoop and Spark, to analyze the Enhanced Health Insurance Claims dataset. The objective is to identify claim patterns, predict high-cost claims, detect anomalies, and generate actionable insights that can support decision-making in the insurance domain.

Suggested Solution

To address these challenges, we propose a distributed big data processing and analytics pipeline using the Hadoop ecosystem:

1. Data Flow & Workflow

Data Ingestion: Upload the dataset into the Hadoop Distributed File System (HDFS) for distributed and fault-tolerant storage.

Preprocessing & Cleaning: Use Spark batch jobs to handle missing values, encode categorical data, normalize numerical fields, and perform feature engineering.

2. Analytical Modules:

Regression/Classification: Predict high-cost claims based on patient, provider, and claim attributes.

Aggregations: Perform large-scale trend analysis (e.g., total claim costs by provider or procedure type).

Clustering/Segmentation: Profile patients and providers to identify behavioral groups or risk categories.

Anomaly Detection: Detect fraudulent or suspicious claims by analyzing unusual patterns.

Visualization & Insights: Build dashboards using Python visualization libraries, Tableau, or Power BI to communicate findings interactively.

3. Scalability & Performance

Hadoop and Spark ensure that the system can scale efficiently to larger datasets.

Parallel processing enables faster computation compared to traditional tools, making real-time or near-real-time insights feasible.

This end-to-end pipeline demonstrates how big data management tools can store, preprocess, and analyze large-scale health insurance data while generating predictive and descriptive insights.

Dataset Description

The Enhanced Health Insurance Claims dataset includes 4,500 synthetic records designed to simulate real-world health insurance claims.

- Rich Features: Patient demographics, claim IDs, provider details, medical procedures, claim costs, and payment outcomes.
- Realistic Scenarios: Captures complexity similar to actual insurance claims, while remaining synthetic and free of privacy risks.
- Analytics-Ready: Suitable for regression, classification, clustering, anomaly detection, and trend analysis.
- Machine Learning Applications: Enables predictive modeling for cost estimation, fraud detection, and patient/provider profiling.

By applying Hadoop and Spark to this dataset, we will demonstrate efficient big data storage, distributed computation, and advanced analytics—highlighting how modern big data frameworks support the insurance sector's need for accuracy, scalability, and actionable insights.

Open source link to the dataset:

<https://www.kaggle.com/datasets/leandrenash/enhanced-health-insurance-claims-dataset>